

Luis A. Rodriguez Condit

Technical Architect – Embedded & Web Systems

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Location: Dallas, TX

Summary

Lead Software Engineer with over 20 years of full-stack experience, specializing in bridging complex, low-level systems with modern web applications used by millions. Knowledgeable in JavaScript, Angular, Java, and C/C++, with proficiency in using WebAssembly and JavaScript to implement core functionality, including calculator emulators in the browser. Proven technical leader, skilled in defining technical requirements, driving code quality, and mentoring teams to deliver robust, performant, and user-centric solutions.

Core Technical Skills

Languages & Frameworks	TypeScript, JavaScript (ES6+), Angular (SPA/Component Architecture), C/C++, Java (OOD), z80/eZ80 Assembly
Frontend & UI/UX	WebAssembly, HTML5, CSS, Cross-Browser/Platform UI, WebUSB/WebSerial Integration, Accessibility (WCAG)
Systems & Low-Level	Processor Emulation (Z80, eZ80, T4, Lapis MCU), Embedded Systems, Control Systems, RTLinux, Communication Protocols
DevOps & Testing	GIT, CI/CD Pipelines (Jenkins/Gulp/CMake), Build Automation, Mocha, Selenium, Jira, Confluence
Architecture & Backend	RESTful Services, Microservices (Analogous Experience), Node.js (Prototyping/Testing), Object-Oriented Design (OOD), JNI
Leadership	Technical Leadership, Scrum Master, Agile/Scrum, Test-Driven Development (TDD)

Professional Experience

Texas Instruments -- Software Architect -- (2010 – 2025, Dallas, TX)

Progressive tenure specializing in bridging low-level hardware emulation and proprietary communication protocols with desktop and web applications. Recognized with an appointment to the first rung of TI's Technical Ladder.

Web Apps

- Architected and led the development of several Single Page Applications (SPAs) using TypeScript and Angular, serving a global user base of over 3.5 million.
- Defined the front-end architecture for multiple iterations of calculator-related SPAs, ensuring responsiveness, and multi-platform compatible.
- Partnered with Product Management and UX/Visual Designers to translate complex requirements into seamless, engaging user experiences.
- Designed and consumed RESTful services and internal APIs to facilitate data transfer, firmware updates, and cloud-based content access between the Angular frontend and backend infrastructure.

Hardware Emulation

- Spearheaded the design and implementation of high-performance calculator emulators for the browser, developing in pure JavaScript (maximizing browser support) for ASICs that use Z80, eZ80, Toshiba T4, and Lapis MCUs.
- Led the adoption of WebAssembly for one major calculator platform, cross-compiling C code to execute natively in the browser.
- Implemented code needed in both z80 assembly and JavaScript to make calculator emulators interact correctly with screen readers (Accessibility).

Communications

- Led the integration of WebUSB and WebSerial APIs into web applications (e.g., TI-Nspire Connect), enabling direct, stable communication between the browser and physical calculators.
- Enabled test automation for calculator emulators through a WebSocket-driven harness that allowed automated tests (Robot Framework + Python) from other platforms to be reused.

Proof of Concept Development and Prototypes

- Championed the adoption of new technologies by delivering fast code implementations and rapid prototypes (e.g., Node.js with Mocha and Selenium for automation entry points).
- Significantly enhanced development efficiency by using Jenkins parallelization, Gulp, and CMake, resulting in a 70% reduction in build times.
- Drove a culture of code quality through code reviews and mentorship, raising the technical bar for junior and mid-level engineers across multiple feature teams.

Digital Chocolate -- Game Engineering Manager -- (2009 – 2010)

- Managed teams responsible for porting mobile games across various US carrier devices, focusing on technical quality and resource allocation. Mentored the engineering team through hands-on tutorials and code deep-dives, directly contributing to team skill development.
- Created a client module for Qualcomm's Application Value Billing API (C/C++) to enable in-app purchases and promote code reuse across future game titles.

LemonQuest -- iPhone Programmer -- (2008 – 2009)

- Ported games from DirectX to iPhone's OpenGL ES, optimizing the user experience by translating desktop inputs for touch and accelerometer use.

Gameloft -- Game Programmer -- (2005 – 2008)

- Led the team that developed the first successful iPhone game prototype for the studio, demonstrating rapid feature development and adaptability to emerging mobile/browser platforms.
- Led the technical effort to port multiple game titles to resource-limited, 3D-capable mobile devices using Java/J2ME and OpenGL ES, focusing on maintaining graphic fidelity and stability within strict memory constraints.
- Directed and participated in team efforts for porting games to severely space-constrained devices (Virgin Mobile catalog). This required strategic scope reduction, asset compression, and intensive micro-optimizations to ensure acceptable performance and quality within limited runtime memory.

Education & Professional Development

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)

- Completed coursework towards: Master's in Electronics and Telecommunications

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)

- Bachelor's in Electronics Systems Engineering

Professional Development: Online Courses

- In Progress: Self-study and projects focused on Docker (see GitHub profile) and modern containerization concepts (Q4 2025).
- Selected courses: iOS Networking & Persistence, UIKit, JavaScript Design Patterns, Web Performance, Software Testing, AI/Robotics.

DISCLAIMER: Portions of this document were created using an AI to tailor experience points to the target role. All content, including dates, titles, and technical claims, is 100% accurate and based on my verified career history. I have thoroughly proofread and confirmed its factual correctness.